Building a Phishing URL Detector

- Legitimate URLS
  - Safe websites links used for comparison

- Phishing URLs
  - Potentially malicious links collected from OpenPhish

URL Feature Analysis

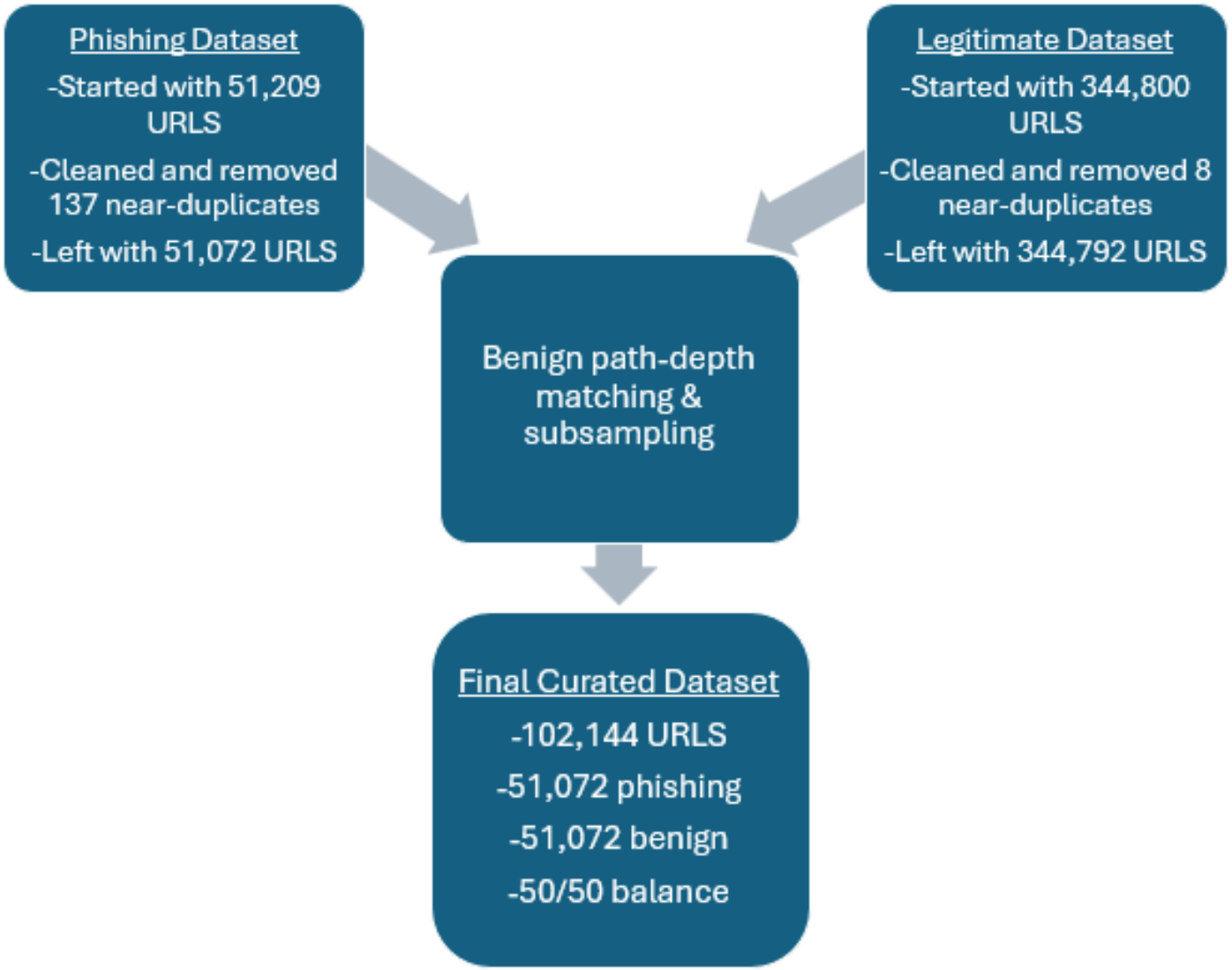
- Length – Digits – Dots – Hyphens – IP Address

Phishing Detection Model

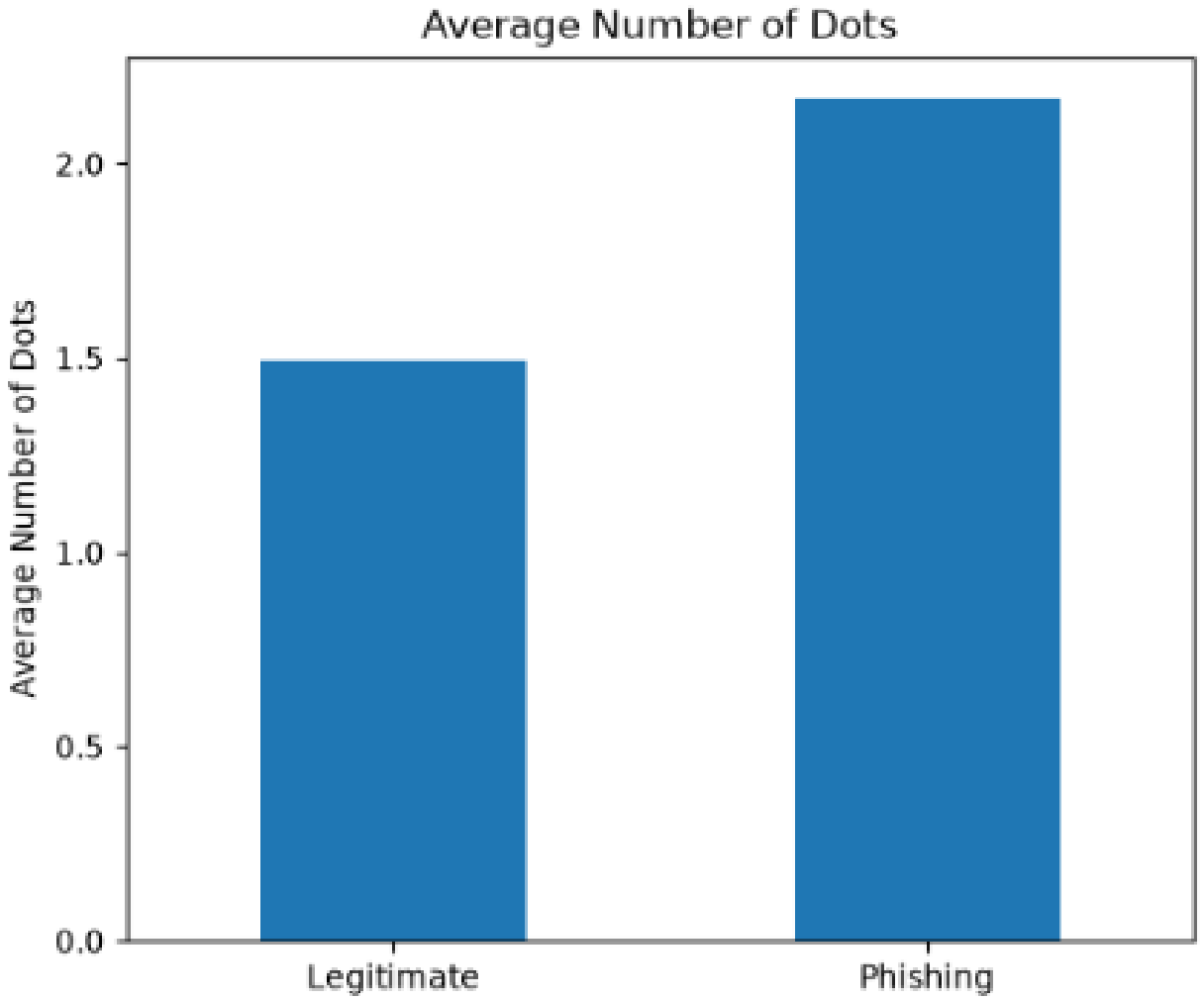
-Learn patterns to distinguish phishing from legitimate URLs

Sydnie Dailey, Katherine Vykhovanets, Ana Madrigal – Fall 2026, Dr. Qingli Zeng

Data


Phishing Dataset
-Started with 51,209 URLS
-Cleaned and removed 137 near-duplicates
-Left with 51,072 URLS
Legitimate Dataset
-Started with 344,800 URLS
-Cleaned and removed 8 near-duplicates
-Left with 344,792 URLS
Benign path-depth matching & subsampling
Final Curated Dataset
-102,144 URLS
-51,072 phishing
-51,072 benign
-50/50 balance

EDA Findings



3-Way Evaluation Table

Split	Precision	Recall	F1	ROC-AUC	PR-AUC	Recall @ 1% FPR	Recall @ 0.1% FPR
Random stratified	0.996004	0.829646	0.905245	0.980961	0.984864	0.884570	0.6494635
Temporal	0.503546	0.653353	0.568750	0.508718	0.508386	0.010181	0.001860
Domain-grouped	0.997022	0.820204	0.900011	0.983547	0.986737	0.893788	0.607290

FPR	Recall	Alarms per Day	P (Phishing | Alarm)
1%	0.884570	500392.285	0.0008838765369853774
0.1%	0.6494635	50319.73175	0.006453368066692843

Lessons Learned and Limitations

- What we learned
  - Phishing and legitimate URLs can share similar characteristics, making phishing detection challenging.
- Model Limitations
  - Simple URL features can cause false positives, and the legitimate dataset may not reflect real browsing activity.
- Key takeaway: Reliable phishing detection requires multiple features and realistic, up to-date datasets